STATISTICS · SAMPLING AND EXPERIMENTATION

Random Assignment Lab

Explore & Reflect TEACHER KEY

Name _____ Date _____ Period _____

Open **Random Assignment Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Students choose whether to use a study app, and stronger students tend to choose it. If the app does nothing, what might the experiment show? Why?
   A fake positive effect: the app group has higher GPAs, so it scores higher anyway. GPA is confounded with using the app.
2. Does random assignment guarantee that the two groups have the same mean GPA?
   No. It balances the groups on average, but any one assignment can be unbalanced by chance.

2 Explore use the tool

1. True effect 0, Students choose, 200+ repetitions: record the center and SD of the differences. What do you conclude?
   Center ≈ +8.6, SD ≈ 1.7. The app looks effective although it does nothing (confounding).
2. Random assignment, true effect 0, 200+ repetitions: record the center and SD.
   Center ≈ 0, SD ≈ 3.0. Random assignment is unbiased.
3. Random assignment: assign once at a time. How many tries until the app group's mean GPA is at least 0.3 higher? What does that show?
   Answers vary; about 1 assignment in 7 (14%). Random assignment balances on average, not every time.
4. Same true effect: compare random assignment with randomized pairs (200+ each). Record both SDs and explain the difference.
   ≈ 3.0 vs. ≈ 1.4. Pairing students with similar GPAs removes most of the GPA-to-GPA variation.

3 Record use the tool and fill in the table

Assignment methods (true effect +4)

200 or more repetitions each. Values vary a little; the key gives long-run values.

Assignment	Center of differences	SD	Unbiased?
Random assignment	≈ +4	≈ 3.0	yes
Randomized pairs	≈ +4	≈ 1.4	yes
Students choose	≈ +12.6	≈ 1.7	no (confounded)

Randomization test: how often is p < 0.05?

Run 10 experiments (1,000 re-randomizations each) for each true effect.

True effect	0	+4	+8	+12
Experiments with p < 0.05 (out of 10)	0 or 1	about 3–4	about 8	about 10
Long-run share	≈ 0.06	≈ 0.36	≈ 0.82	≈ 0.99

4 Explain what did you discover?

1. When students choose their group, GPA is **confounded** with the treatment, so we can't tell which one caused the difference.
2. Random assignment makes the groups similar **on average (in the long run)**, not in every single experiment.
3. Randomizing within matched pairs **reduces** the variability of the estimates.
4. A randomization-test p-value is the share of shuffles with a difference **at least as large as** the observed one, assuming **the app has no effect**.
5. A randomization test for the app gives a p-value of 0.31. Explain what this number means in context, and what you can conclude.
   If the app had no effect, about 31% of random reassignments of these 20 scores would give a difference at least as large as the one observed. That is common, so the result is not convincing evidence that the app raises scores (it does not prove the app has no effect).

5 Apply a new situation, no tool

1. Students who chose to attend tutoring had higher grades than students who didn't. Why can't we conclude that tutoring caused the higher grades? Describe an experiment that could.
   Students who choose tutoring may be more motivated (confounding). Randomly assign volunteers to tutoring or no tutoring, then compare grades.
2. Identify the design: 30 pairs of twins; in each pair a coin flip decides which twin gets the new program.
   A matched-pairs (randomized block) design.
3. In a randomization test with 500 shuffles, 9 differences were at least as large as the observed one. Find the p-value and state a conclusion at $\alpha = 0.05$.
   $\frac{9}{500} = 0.018 < 0.05$: convincing evidence of a real effect.

Teacher notes

Intended discovery: Random assignment balances lurking variables on average, so differences center on the true effect, while self-selection confounds the treatment with GPA; matching reduces variability; a randomization test's p-value is the share of label shuffles at least as extreme as the observed difference.

Likely misconceptions:

- Random assignment guarantees identical groups.
- Any observed difference proves the treatment works.
- A large p-value proves the treatment has no effect.

Answers that can vary: All simulated values; the number of tries in Explore 3; the counts out of 10 in the second Record table (long-run shares are given).